

# WHEN DOES A BOX FIT INTO A BOX? AN EXPLICIT CRITERION WITH A FORMAL PROOF

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**ABSTRACT.** We give a necessary and sufficient condition, in closed form, for a rectangular box with edges  $q_1, q_2, q_3 \geq 0$  to fit, after an arbitrary rigid motion, into a rectangular box with edges  $p_1, p_2, p_3 > 0$ . The condition is a disjunction of nine explicit checks, each consisting of two planar tests expressed by a formula with a single square root. The key geometric fact is that if a box fits at all, it fits in a position where one of its edges is perpendicular to one of the axes of the container; we prove this by showing that at any other position all three widths of the box can be decreased simultaneously. The problem was posed by Garnett in 1923 and was described by Jerrard and Wetzel in 2004 as unsolved. We also give the criterion for strict fitting (without touching the container), which is not obtained by merely making the inequalities strict. The results have been verified in the Lean 4 proof assistant with the Mathlib library.

## 1. INTRODUCTION

Let  $P = [0, p_1] \times [0, p_2] \times [0, p_3]$  with  $p_i > 0$  and  $Q = [0, q_1] \times [0, q_2] \times [0, q_3]$  with  $q_j \geq 0$ . We say that  $Q$  *fits into*  $P$  if  $f(Q) \subseteq P$  for some isometry  $f$  of  $\mathbb{R}^3$ . The question of when this happens was posed by F. M. Garnett in 1923 [4]; W. B. Carver [2] gave a partial answer, assuming without proof that in an optimal position all vertices of  $Q$  lie on faces of  $P$ . The planar analogue, a rectangle in a rectangle, was settled by Carver in 1957 [3] (see also Wetzel [13]). Jerrard and Wetzel [5, p. 22] describe the three-dimensional question as an instance of a “general unsolved fitting problem for boxes in higher dimensions”. Among related results we mention the treatment of a disk in a box [1], of a circular cylinder along the space diagonal of a box [6], where a criterion “along the lines of Post” [10] is asked for and described as unsolved, and the containment algorithms of Martin and Stephenson [8]. The olympiad problem [7] proves the necessary condition  $q_1 + q_2 + q_3 \leq p_1 + p_2 + p_3$ .

In this paper we give such a criterion for boxes. It is a disjunction of explicit checks, much as Post’s criterion for a triangle in a triangle [10] is a disjunction of 18 inequalities.

**The criterion.** For  $x \geq y \geq 0$  and  $s \geq 0$  put

$$(1) \quad H(x, y; s) = \begin{cases} y, & x \leq s, \\ \min\left(x, \frac{2xys + (x^2 - y^2)\sqrt{x^2 + y^2 - s^2}}{x^2 + y^2}\right), & x > s, \end{cases}$$

and for arbitrary  $u, v \geq 0$  let  $H(u, v; s) = H(\max(u, v), \min(u, v); s)$ . By Lemma 3.1, when  $\min(u, v) \leq s$  the number  $H(u, v; s)$  is the least possible extent along a strip of width  $s$  of a  $u \times v$  rectangle placed in the strip. For  $a, b, u, v \geq 0$  put

$$\text{Fit}(a, b; u, v) :\iff \min(u, v) \leq a \text{ and } H(u, v; a) \leq b.$$

This says that the  $u \times v$  rectangle fits into the  $a \times b$  rectangle (Remark 3.2).

**Definition 1.1** (Criterion NC). Let  $(i, i', i'')$  be a permutation of the axes  $(1, 2, 3)$  of  $P$  and  $(j, k, l)$  a permutation of the edges  $(1, 2, 3)$  of  $Q$ . The *check*  $\text{NC}_{(i, i', i''), (j, k, l)}$  is

$$\min(q_k, q_l) \leq p_i \quad \text{and} \quad \text{Fit}(p_{i'}, p_{i''}; q_j, h^*), \quad h^* := H(q_k, q_l; p_i).$$

$\text{NC}(p, q)$  holds if at least one check holds. Since  $H$  is symmetric in its first two arguments, there are 9 choices of  $(i, j)$  and two orders of the “floor” axes  $(i', i'')$  for each.

Geometrically, the check describes positions in which edge  $j$  of  $Q$  is horizontal, i.e. perpendicular to axis  $i$  of  $P$  (the “height”). The face  $q_k \times q_l$  then stands in the vertical strip of height  $p_i$ ; its trace on the floor is a segment of length at least  $h^*$ , and the rectangle  $q_j \times h^*$  must fit into the floor  $p_{i'} \times p_{i''}$ .

**Theorem 1.2** (Main theorem). *Let  $p_1, p_2, p_3 > 0$  and  $q_1, q_2, q_3 \geq 0$ . The box  $Q$  fits into the box  $P$  if and only if  $\text{NC}(p, q)$  holds.*

Sufficiency is constructive (Section 4): every check that holds yields an explicit rotation. Necessity rests on the following theorem (Section 6), which is the main new ingredient. Here  $w_i(R) = \sum_j q_j |R_{ij}|$  is the width of the rotated box along axis  $i$  (Section 2).

**Theorem 1.3** (Descent). *Let  $q_1, q_2, q_3 > 0$  and let  $R \in \text{O}(3)$  have no zero entries. Then there is  $R' \in \text{O}(3)$  with  $w_i(R') < w_i(R)$  for  $i = 1, 2, 3$ .*

**Formal verification.** Theorem 1.2 has been formalized and checked in Lean 4 with Mathlib; see Section 10. The formal statement uses literally the formulas above; the only definitions on which its meaning depends are those of a box and of fitting.

## 2. REDUCTION TO WIDTHS

Every isometry of  $\mathbb{R}^3$  has the form  $x \mapsto Rx + t$  with  $R \in \text{O}(3)$  (Mazur–Ulam). Column  $j$  of  $R$  is the direction of edge  $j$  of the moved box, and  $R_{ij}$  is the cosine of the angle between this edge and axis  $i$  of  $P$ .

**Lemma 2.1.** *Let  $R$  be any real  $3 \times 3$  matrix. Some translate of  $RQ$  lies in  $P$  if and only if*

$$w_i(R) := \sum_{j=1}^3 q_j |R_{ij}| \leq p_i \quad (i = 1, 2, 3).$$

*Consequently  $Q$  fits into  $P$  if and only if  $w(R) \leq p$  for some  $R \in \text{O}(3)$ .*

*Proof.* The  $i$ -th coordinate of  $Rx = \sum_j R_{ij}x_j$ ,  $x \in Q$ , ranges over an interval of length  $\sum_j q_j |R_{ij}|$ , the endpoints being attained at vertices of  $Q$ . The translations along different axes are independent.  $\square$

The same argument in the plane shows that a  $u \times v$  rectangle fits into an  $a \times b$  rectangle if and only if  $uc + vd \leq a$  and  $ud + vc \leq b$  for some  $c, d \geq 0$  with  $c^2 + d^2 = 1$ .

## 3. A RECTANGLE IN A STRIP

Let  $x \geq y \geq 0$  and  $s \geq 0$ . For  $c, d \geq 0$  with  $c^2 + d^2 = 1$  put  $A = xc + yd$  (the extent of the rotated  $x \times y$  rectangle across the strip) and  $B = xd + yc$  (its extent along the strip).

**Lemma 3.1** (Strip lemma). (a) *If  $y \leq s$ , there are  $c, d \geq 0$  with  $c^2 + d^2 = 1$ ,  $A \leq s$  and  $B \leq H(x, y; s)$ .*

(b) *If  $c, d \geq 0$ ,  $c^2 + d^2 = 1$  and  $A \leq s$ , then  $y \leq s$  and  $H(x, y; s) \leq B$ .*

*The same holds for arbitrary  $u, v \geq 0$  in place of  $x, y$  with  $H(u, v; s)$ ,  $A = uc + vd$ ,  $B = ud + vc$  (exchange  $c$  and  $d$  if  $u < v$ ).*

*Proof.* If  $x = 0$  everything is trivial, so let  $r = \sqrt{x^2 + y^2} > 0$  and let  $\alpha \in [0, \pi/4]$  be given by  $\cos \alpha = x/r$ ,  $\sin \alpha = y/r$ . Write  $(c, d) = (\cos \theta, \sin \theta)$  with  $\theta \in [0, \pi/2]$ . Then

$$A(\theta) = r \cos(\theta - \alpha), \quad B(\theta) = r \sin(\theta + \alpha).$$

For  $\theta \in [0, \pi/2]$  we have  $\theta - \alpha \in [-\pi/2, \pi/2]$  and  $\theta + \alpha \in [0, \pi]$ , so both  $A$  and  $B$  are concave on  $[0, \pi/2]$ ; hence on every subinterval they attain their minima at the endpoints. Moreover  $A(0) = B(\pi/2) = x$  and  $A(\pi/2) = B(0) = y$ .

By concavity  $A \geq \min(x, y) = y$ , which proves the first claim of (b). If  $x \leq s$ , the angle  $\theta = 0$  is admissible with  $B(0) = y$ , and  $B \geq \min(B(0), B(\pi/2)) = y$  everywhere; this proves both (a) and (b).

Let  $y \leq s < x$ . On  $[0, \alpha]$  the function  $A$  increases from  $x$  to  $r$ , so  $A > s$  there; on  $[\alpha, \pi/2]$  it decreases from  $r$  to  $y \leq s$ . Hence the admissible angles form an interval  $[\theta^*, \pi/2]$  with  $A(\theta^*) = s$ , and the minimum of the concave function  $B$  over it equals  $\min(B(\theta^*), B(\pi/2)) = \min(B(\theta^*), x)$ . It remains to compute  $B(\theta^*)$ . With  $\varphi = \theta^* - \alpha \in [0, \pi/2]$  we have  $\cos \varphi = s/r$ ,  $\sin \varphi = \sqrt{r^2 - s^2}/r$ , and

$$\cos \theta^* = \frac{xs - y\sqrt{r^2 - s^2}}{r^2}, \quad \sin \theta^* = \frac{ys + x\sqrt{r^2 - s^2}}{r^2}, \quad B(\theta^*) = \frac{2xys + (x^2 - y^2)\sqrt{r^2 - s^2}}{r^2}.$$

Thus  $\min\{B : A \leq s\} = H(x, y; s)$ , which gives (a) (take the minimizing endpoint) and (b).  $\square$

*Remark 3.2.* By the planar version of Lemma 2.1 and Lemma 3.1,  $\text{Fit}(a, b; u, v)$  holds if and only if the  $u \times v$  rectangle fits into the  $a \times b$  rectangle. Carver [3] gave another closed form of the same condition: for  $u \geq v$ ,  $a \geq b$ ,  $u > a$ ,  $v \leq b$  it reads  $\left(\frac{a+b}{u+v}\right)^2 + \left(\frac{a-b}{u-v}\right)^2 \geq 2$ . We do not use Carver's form. (Numerically the two forms agree on  $2 \cdot 10^5$  random rectangles; see `tools/box_fit.py`.)

#### 4. SUFFICIENCY

**Proposition 4.1.** *If a check of  $\text{NC}(p, q)$  holds, then  $Q$  fits into  $P$ , and an explicit rotation can be written down.*

*Proof.* Relabeling axes and edges (which permutes rows and columns of  $R$ ) we may assume that the check is  $\text{NC}_{(1,2,3),(1,2,3)}$ :  $\min(q_2, q_3) \leq p_1$  and  $\text{Fit}(p_2, p_3; q_1, h^*)$  with  $h^* = H(q_2, q_3; p_1)$ . By Lemma 3.1(a) there are  $c_1, d_1 \geq 0$ ,  $c_1^2 + d_1^2 = 1$ , with

$$q_2 c_1 + q_3 d_1 \leq p_1, \quad q_2 d_1 + q_3 c_1 \leq h^*,$$

in particular  $h^* \geq 0$ . Applying Lemma 3.1(a) to the rectangle  $q_1 \times h^*$  in the strip of width  $p_2$  gives  $c_2, d_2 \geq 0$ ,  $c_2^2 + d_2^2 = 1$ , with

$$q_1 c_2 + h^* d_2 \leq p_2, \quad q_1 d_2 + h^* c_2 \leq H(q_1, h^*; p_2) \leq p_3.$$

Let

$$R = \begin{pmatrix} 0 & c_1 & -d_1 \\ c_2 & -d_1 d_2 & -c_1 d_2 \\ d_2 & d_1 c_2 & c_1 c_2 \end{pmatrix}.$$

Its columns  $f_1 = (0, c_2, d_2)$ ,  $f_2 = c_1 e_1 + d_1 n$ ,  $f_3 = -d_1 e_1 + c_1 n$  with  $n = (0, -d_2, c_2)$  are orthonormal, so  $R \in \text{O}(3)$ . Its widths are

$$\begin{aligned} w_1 &= q_2 c_1 + q_3 d_1 \leq p_1, \\ w_2 &= q_1 c_2 + d_2 (q_2 d_1 + q_3 c_1) \leq q_1 c_2 + d_2 h^* \leq p_2, \\ w_3 &= q_1 d_2 + c_2 (q_2 d_1 + q_3 c_1) \leq q_1 d_2 + c_2 h^* \leq p_3, \end{aligned}$$

and Lemma 2.1 concludes the proof.  $\square$

## 5. POSITIONS WITH A HORIZONTAL EDGE

**Proposition 5.1.** *Let  $q \geq 0$ ,  $p > 0$ , and let  $R \in O(3)$  satisfy  $w(R) \leq p$  and  $R_{ij} = 0$  for some  $i, j$ . Then  $\text{NC}(p, q)$  holds.*

*Proof.* Relabeling, let  $R_{11} = 0$ . For an orthogonal matrix  $R = (\det R) \text{cof}(R)$  and  $\det R = \pm 1$ , so each  $|R_{ab}|$  equals the absolute value of the corresponding cofactor. Since  $R_{11} = 0$ , this gives

$$|R_{22}| = |R_{13}||R_{31}|, \quad |R_{23}| = |R_{12}||R_{31}|, \quad |R_{32}| = |R_{13}||R_{21}|, \quad |R_{33}| = |R_{12}||R_{21}|.$$

Put  $c = |R_{12}|$ ,  $d = |R_{13}|$ ,  $c' = |R_{21}|$ ,  $d' = |R_{31}|$ ; then  $c^2 + d^2 = 1$  (first row) and  $c'^2 + d'^2 = 1$  (first column). With  $B := q_2 d + q_3 c$  we obtain

$$w_1 = q_2 c + q_3 d, \quad w_2 = q_1 c' + d' B, \quad w_3 = q_1 d' + c' B.$$

From  $w_1 \leq p_1$  and Lemma 3.1(b),  $\min(q_2, q_3) \leq p_1$  and  $h^* := H(q_2, q_3; p_1) \leq B$ ; also  $h^* \geq 0$  by Lemma 3.1(a). Hence  $q_1 c' + d' h^* \leq w_2 \leq p_2$  and  $q_1 d' + c' h^* \leq w_3 \leq p_3$ . Lemma 3.1(b), applied to the rectangle  $q_1 \times h^*$  in the strip of width  $p_2$  with direction  $(c', d')$ , gives  $\min(q_1, h^*) \leq p_2$  and  $H(q_1, h^*; p_2) \leq q_1 d' + h^* c' \leq p_3$ , i.e.  $\text{Fit}(p_2, p_3; q_1, h^*)$ . Thus  $\text{NC}_{(1,2,3),(1,2,3)}$  holds.  $\square$

## 6. DESCENT: PROOF OF THEOREM 1.3

Throughout this section  $q_1, q_2, q_3 > 0$  and  $R \in O(3)$  has no zero entries. We write  $e_1, e_2, e_3$  for the standard basis,  $u \cdot v$  for the scalar product and  $u \times v$  for the vector product.

**6.1. Linearization.** Let  $s_{ij} = \text{sgn } R_{ij} \in \{\pm 1\}$ ,  $g_i = (s_{i1}q_1, s_{i2}q_2, s_{i3}q_3)$  and  $a_i = Rg_i \in \mathbb{R}^3$ . Write  $A_{ik} = (a_i)_k$ . Then  $A_{ii} = \sum_j R_{ij}s_{ij}q_j = w_i(R)$ . If  $C \in O(3)$  and  $CR$  has the same sign pattern as  $R$ , then

$$(2) \quad w_i(CR) = \sum_j q_j s_{ij} (CR)_{ij} = (Ca_i)_i.$$

**Lemma 6.1** (B1).  $|A_{ik}| < A_{kk}$  for  $i \neq k$ .

*Proof.*  $|A_{ik}| = |\sum_j s_{ij}q_j R_{kj}| \leq \sum_j q_j |R_{kj}| = A_{kk}$ . Equality would force all  $s_{ij}R_{kj}$  ( $q_j > 0$ ,  $R_{kj} \neq 0$ ) to have the same sign, hence all products  $R_{ij}R_{kj}$  to be nonzero of the same sign, contradicting the orthogonality of rows  $i$  and  $k$ .  $\square$

**6.2. The Cayley rotation.** For  $v \in \mathbb{R}^3$  let  $[v]_\times x = v \times x$  and

$$C(v) = \frac{(1 - |v|^2)I + 2[v]_\times + 2vv^\top}{1 + |v|^2}.$$

This is the rotation given by the quaternion  $(1, v)$ ; one checks directly that  $C(v)C(v)^\top = I$ . Moreover  $C(0) = I$  and  $C$  is continuous. For  $a \in \mathbb{R}^3$  put

$$b_i(a) = a \times e_i, \quad P_i(a; v) = v_i(v \cdot a) - |v|^2 a_i, \quad D_i(a; v) = b_i(a) \cdot v + P_i(a; v).$$

Since  $e_i \cdot (v \times a) = v \cdot (a \times e_i)$ , a direct computation gives

$$(3) \quad (C(v)a)_i - a_i = \frac{2}{1 + |v|^2} D_i(a; v).$$

Note that  $P_i(a; \cdot)$  is a quadratic form:  $P_i(a; tv) = t^2 P_i(a; v)$ .

**6.3. Descent along a curve.** We shall rotate along  $v(t) = t(h + tz)$ ,  $t > 0$  small. Then

$$(4) \quad D_i(a; v(t)) = t b_i(a) \cdot h + t^2 (b_i(a) \cdot z + P_i(a; h + tz)).$$

**Lemma 6.2.** *Let  $a, h, z \in \mathbb{R}^3$  and  $b = b_i(a)$ .*

- (a) *If  $b \cdot h < 0$ , or  $b \cdot h = 0$  and  $b \cdot z + P_i(a; h) < 0$ , then  $D_i(a; v(t)) < 0$  for all sufficiently small  $t > 0$ .*
- (b) *If  $b = 0$ ,  $a_i > 0$  and all components of  $z$  are nonzero, then  $D_i(a; v(t)) < 0$  for all sufficiently small  $t > 0$ .*

*Proof.* (a) By (4),  $D_i/t = b \cdot h + t(b \cdot z + P_i(a; h + tz)) \rightarrow b \cdot h$  as  $t \rightarrow 0^+$ ; in the second case  $D_i/t^2 = b \cdot z + P_i(a; h + tz) \rightarrow b \cdot z + P_i(a; h)$ . (b)  $b = a \times e_i = 0$  means  $a = a_i e_i$ ; then  $P_i(a; u) = u_i^2 a_i - |u|^2 a_i = -a_i \sum_{m \neq i} u_m^2$ . For small  $t > 0$  every component  $h_m + tz_m$  of  $u = h + tz$  is nonzero (if  $h_m \neq 0$  by continuity, if  $h_m = 0$  because  $z_m \neq 0$ ). Hence  $D_i(a; v(t)) = t^2 P_i(a; h + tz) < 0$ .  $\square$

#### 6.4. The key inequality.

**Lemma 6.3.** *Let  $a_1, a_2, a_3 \in \mathbb{R}^3$  satisfy  $|A_{ik}| < A_{kk}$  for  $i \neq k$ , where  $A_{ik} = (a_i)_k$ . Let  $\nu \in \mathbb{R}^3$ ,  $\nu \geq 0$ , have at least two positive entries and satisfy  $\sum_i \nu_i b_i(a_i) = 0$ . Then  $\sum_i \nu_i P_i(a_i; h) < 0$  for every  $h \neq 0$ .*

*Proof.* Since  $a \times e_1 = (0, a_3, -a_2)$ ,  $a \times e_2 = (-a_3, 0, a_1)$ ,  $a \times e_3 = (a_2, -a_1, 0)$ , the condition  $\sum_i \nu_i b_i(a_i) = 0$  says precisely that

$$\nu_i A_{ik} = \nu_k A_{ki} \quad (i \neq k).$$

Put  $d_i = \nu_i A_{ii} \geq 0$  and  $m_{ik} = \nu_i A_{ik} = \nu_k A_{ki}$ . Expanding and using the symmetry,

$$\sum_i \nu_i P_i(a_i; h) = \sum_i \nu_i \left( h_i \sum_k h_k A_{ik} - |h|^2 A_{ii} \right) = - \sum_{i < k} B_{ik}, \quad B_{ik} = d_i h_k^2 + d_k h_i^2 - 2m_{ik} h_i h_k.$$

By the hypothesis,  $m_{ik}^2 = \nu_i \nu_k A_{ik} A_{ki} \leq \nu_i \nu_k |A_{ik}| |A_{ki}| \leq \nu_i \nu_k A_{kk} A_{ii} = d_i d_k$ , with strict inequality if  $\nu_i \nu_k > 0$ .

Each  $B_{ik} \geq 0$ : if  $d_i = 0$  then  $m_{ik} = 0$  and  $B_{ik} = d_k h_i^2$ ; if  $d_i > 0$  then  $d_i B_{ik} = (d_i h_k - m_{ik} h_i)^2 + (d_i d_k - m_{ik}^2) h_i^2 \geq 0$ .

Some  $B_{ik}$  is positive. Choose  $c$  with  $h_c \neq 0$  and, using the two positive entries of  $\nu$ , an index  $i \neq c$  with  $\nu_i > 0$ , so  $d_i > 0$ . If  $d_c = 0$  then  $m_{ic} = 0$  and  $B_{ic} = d_i h_c^2 > 0$ . If  $d_c > 0$  then  $\nu_c > 0$ , hence  $m_{ic}^2 < d_i d_c$ , and  $d_i B_{ic} = (d_i h_c - m_{ic} h_i)^2 + (d_i d_c - m_{ic}^2) h_i^2$  is positive both for  $h_i \neq 0$  and for  $h_i = 0$ .  $\square$

**6.5. Choice of the curve.** We use Motzkin's theorem of the alternative [9, 11] in the following form: for vectors  $b_i \in \mathbb{R}^3$  and numbers  $c_i$ ,  $i \in K$ , either the system  $b_i \cdot z < c_i$  ( $i \in K$ ) has a solution, or there is  $\mu \geq 0$ ,  $\mu \neq 0$ , supported on  $K$ , with  $\sum_i \mu_i b_i = 0$  and  $\sum_i \mu_i c_i \leq 0$ .

**Lemma 6.4.** *Let  $a_i$  be as in the Linearization,  $b_i = b_i(a_i)$  and  $K = \{i : b_i \neq 0\}$ . There are  $h, z \in \mathbb{R}^3$  such that all components of  $z$  are nonzero and, for each  $i \in K$ , either  $b_i \cdot h < 0$ , or  $b_i \cdot h = 0$  and  $b_i \cdot z + P_i(a_i; h) < 0$ .*

*Proof.* Apply Motzkin's theorem with  $c = 0$ . If  $b_i \cdot h < 0$  for all  $i \in K$  has a solution  $h$ , take  $z = (1, 1, 1)$ . Otherwise there is  $\mu \neq 0$ ,  $\mu \geq 0$ , with  $\sum \mu_i b_i = 0$  (note  $b_i = 0$  for  $i \notin K$ ), so the matrix with rows  $b_1, b_2, b_3$  is singular and there is  $h \neq 0$  with  $b_i \cdot h = 0$  for all  $i$ . Apply Motzkin's theorem again, with  $c_i = -P_i(a_i; h)$ .

If there is  $z_0$  with  $b_i \cdot z_0 + P_i(a_i; h) < 0$  for  $i \in K$ , then  $z = z_0 + \delta(1, 1, 1)$  with small  $\delta > 0$  keeps these strict inequalities and has all components nonzero.

Otherwise there is  $\nu \geq 0$ ,  $\nu \neq 0$ , supported on  $K$ , with  $\sum \nu_i b_i = 0$  and  $\sum \nu_i P_i(a_i; h) \geq 0$ . If  $\nu$  had a single positive entry  $\nu_i$ , then  $\nu_i b_i = 0$  would give  $b_i = 0$ , contradicting  $i \in K$ . So  $\nu$  has two positive entries, and Lemma 6.3 (whose hypothesis holds by Lemma 6.1) gives  $\sum \nu_i P_i(a_i; h) < 0$ , a contradiction.  $\square$

**6.6. Proof of Theorem 1.3.** Take  $h, z$  from Lemma 6.4 and put  $R(t) = C(v(t))R$ . For all sufficiently small  $t > 0$ :

- $R(t) \in O(3)$  has the same sign pattern as  $R$ , because  $R(t) \rightarrow R$  and  $R$  has no zero entries;
- $D_i(a_i; v(t)) < 0$  for each  $i$ : for  $i \in K$  by Lemma 6.2(a), and for  $i \notin K$  by Lemma 6.2(b), since  $(a_i)_i = w_i(R) > 0$ .

By (2) and (3),  $w_i(R(t)) - w_i(R) = (C(v(t))a_i)_i - (a_i)_i = \frac{2}{1+|v(t)|^2} D_i(a_i; v(t)) < 0$  for  $i = 1, 2, 3$ .  $\square$

## 7. PROOF OF THE MAIN THEOREM

Sufficiency is Proposition 4.1. For necessity, by Proposition 5.1 it suffices to show: *if  $Q$  fits into  $P$ , then  $w(R) \leq p$  for some  $R \in O(3)$  with a zero entry.*

*Case  $q_1, q_2, q_3 > 0$ .* The function  $F(R) = \max_i w_i(R)/p_i$  is continuous on the compact group  $O(3)$  and attains its minimum at some  $R^*$ ; since  $Q$  fits,  $F(R^*) \leq 1$ , i.e.  $w(R^*) \leq p$ . If  $R^*$  had no zero entries, Theorem 1.3 would give  $R'$  with  $w_i(R') < w_i(R^*)$  for all  $i$ , hence  $F(R') < F(R^*)$ . So  $R^*$  has a zero entry.

*General case  $q \geq 0$ .* Let  $w(R_0) \leq p$  and suppose that no  $R$  in the compact set  $Z = \{R \in O(3) : R_{ij} = 0 \text{ for some } i, j\}$  satisfies  $w(R) \leq p$ . Then the continuous function  $\varphi(R) = \max_i (w_i(R) - p_i)$  is positive on  $Z$  and has a minimum  $m > 0$  there. Let  $\delta = m/6$ ,  $q' = q + \delta(1, 1, 1) > 0$  and  $p' = p + 3\delta(1, 1, 1)$ . Since  $\sum_j |R_{ij}| \leq 3$  for  $R \in O(3)$ , we have  $w_i(R_0; q') = w_i(R_0; q) + \delta \sum_j |(R_0)_{ij}| \leq p'_i$ , so  $q'$  fits into  $p'$ . By the previous case there is  $R_1 \in Z$  with  $w(R_1; q') \leq p'$ , hence  $w_i(R_1; q) \leq w_i(R_1; q') \leq p_i + 3\delta$  and  $\varphi(R_1) \leq 3\delta < m$ , a contradiction.  $\square$

## 8. EXAMPLES

*Example 8.1 (A rod).* Let  $q = (L, 0, 0)$ . Take the check with height  $i = 1$ , horizontal edge  $j = 2$  and  $(k, l) = (1, 3)$ . Then  $h^* = H(L, 0; p_1) = \sqrt{\max(L^2 - p_1^2, 0)}$ , and  $\text{Fit}(p_2, p_3; 0, h^*)$  reduces to  $h^{*2} \leq p_2^2 + p_3^2$ . Thus the check holds if and only if  $L^2 \leq p_1^2 + p_2^2 + p_3^2$ ; conversely, a longer rod cannot fit since  $\sqrt{p_1^2 + p_2^2 + p_3^2}$  is the diameter of  $P$ . The rod fits exactly along the space diagonal, and this position has an edge (of length 0) perpendicular to an axis.

*Example 8.2 (A square in a cube).* The largest square that fits into the unit cube has side  $a = 3\sqrt{2}/4$  (Nieuwland, published by van Swinden [12]). For  $q = (a, a, 0)$  and  $p = (1, 1, 1)$  take  $j = 1$ ,  $(k, l) = (2, 3)$ :  $h^* = H(a, 0; 1) = \sqrt{a^2 - 1} = \sqrt{2}/4$ . For the floor,  $x^2 = 9/8$ ,  $y^2 = 1/8$ ,  $x^2 + y^2 = 5/4$ ,  $\sqrt{x^2 + y^2 - 1} = 1/2$ ,  $2xy = 3/4$ , so the second branch of (1) equals  $(3/4 + 1 \cdot \frac{1}{2})/(5/4) = 1$  and  $H(a, h^*; 1) = \min(a, 1) = 1$ : the check holds with equality, in accordance with Nieuwland's result.

*Example 8.3 (Classical necessary conditions are not sufficient).* For  $p = (1, 1, 1)$  and  $q = (1.1, 1.1, 0.01)$  the sums of edges, of squared edges, the surface areas and the volumes are all smaller for  $Q$ , yet  $Q$  does not fit: its  $1.1 \times 1.1$  face would give a square larger than  $3\sqrt{2}/4$  in the cube.

## 9. STRICT FITTING

We say that  $Q$  fits strictly into  $P$  if  $f(Q)$  lies in the open box  $(0, p_1) \times (0, p_2) \times (0, p_3)$  for some isometry  $f$ , i.e.  $Q$  can be placed without touching the boundary of  $P$ . This holds if and only if  $w(R) < p$  for some  $R \in O(3)$ : as in the proof of Lemma 2.1, the  $i$ -th coordinate of  $RQ$  fills a closed interval of length  $w_i(R)$ , and some translate of it lies in the open interval  $(0, p_i)$  if and only if  $w_i(R) < p_i$ .

**Theorem 9.1.** *Let  $q \geq 0$ . The box  $Q$  fits strictly into  $P$  if and only if there is  $\varepsilon > 0$  with  $\varepsilon < \min_i p_i$  such that  $\text{NC}(p - \varepsilon(1, 1, 1), q)$  holds.*

*Proof.* If  $w(R) < p$ , then  $w(R) \leq p - \varepsilon(1, 1, 1)$  for  $\varepsilon = \frac{1}{2} \min_i (p_i - w_i(R))$ ; apply Theorem 1.2 to  $p - \varepsilon(1, 1, 1)$ . Conversely, Theorem 1.2 gives  $R$  with  $w(R) \leq p - \varepsilon(1, 1, 1) < p$ .  $\square$

An explicit criterion is obtained by making the inequalities strict, but *not only* them. For  $x \geq y \geq 0, s \geq 0$  put

$$H^\circ(x, y; s) = \begin{cases} y, & x < s, \\ \text{the second line of (1)}, & x \geq s, \end{cases}$$

$H^\circ(u, v; s) = H^\circ(\max(u, v), \min(u, v); s)$ , and  $\text{Fit}^\circ(a, b; u, v) : \iff \min(u, v) < a$  and  $H^\circ(u, v; a) < b$ . Let  $\text{NC}^\circ$  be the criterion of Definition 1.1 with  $\min(q_k, q_l) < p_i$ ,  $h^* = H^\circ(q_k, q_l; p_i)$  and  $\text{Fit}^\circ$  in place of  $\min(q_k, q_l) \leq p_i$ ,  $H$  and  $\text{Fit}$ .

**Theorem 9.2.** *Let  $p > 0, q \geq 0$ . The box  $Q$  fits strictly into  $P$  if and only if  $\text{NC}^\circ(p, q)$  holds.*

We use the notation  $A = xc + yd$ ,  $B = xd + yc$  of Section 3, and write  $T(s) = (2xys + (x^2 - y^2)\sqrt{x^2 + y^2 - s^2})/(x^2 + y^2)$  for the second line of (1) without the minimum. Note that  $H^\circ \geq 0$ .

**Lemma 9.3** (Strict strip lemma). *Let  $x \geq y \geq 0$  and  $s \geq 0$ .*

(a $^\circ$ ) *If  $y < s$  and  $H^\circ(x, y; s) < t$ , there are  $c, d \geq 0$  with  $c^2 + d^2 = 1$ ,  $A < s$  and  $B < t$ .*

(b $^\circ$ ) *If  $c, d \geq 0$ ,  $c^2 + d^2 = 1$  and  $A < s$ , then  $y < s$  and  $H^\circ(x, y; s) \leq B$ .*

*The same holds for arbitrary  $u, v \geq 0$  in place of  $x, y$ , as in Lemma 3.1.*

*Proof.* We use the angle  $\theta$ , the functions  $A(\theta)$ ,  $B(\theta)$  and their properties from the proof of Lemma 3.1.

(b $^\circ$ ) By concavity  $A \geq y$ , so  $y < s$ . If  $x < s$ , then  $H^\circ = y \leq \min(B(0), B(\pi/2)) \leq B$ . Let  $x \geq s$ ; then  $x > 0$ . On  $[0, \alpha]$  we have  $A \geq x \geq s$ , so  $\theta \in (\alpha, \pi/2]$ . On  $[\alpha, \pi/2]$  the function  $A$  decreases from  $r \geq x \geq s$  to  $y < s$ , so there is  $\theta^* \in [\alpha, \pi/2]$  with  $A(\theta^*) = s$ , and  $A(\theta) < s$  forces  $\theta > \theta^*$ . The computation in the proof of Lemma 3.1 (valid for every  $s \in [y, r]$ ) gives  $B(\theta^*) = T(s)$ , and by concavity of  $B$  on  $[\theta^*, \pi/2]$  we get  $B(\theta) \geq \min(T(s), x) = H^\circ(x, y; s)$ .

(a $^\circ$ ) If  $x < s$ , take  $\theta = 0$ :  $A = x < s$  and  $B = y = H^\circ < t$ . Let  $x \geq s$ , so  $H^\circ = \min(x, T(s)) < t$ . If  $x < t$ , take  $\theta = \pi/2$ :  $A = y < s$ ,  $B = x < t$ . Otherwise  $T(s) < t$ . Since  $T$  is continuous, there is  $s' \in (y, s)$  with  $T(s') < t$ . As  $y < s' < s \leq x$ , Lemma 3.1(a) applied to the strip of width  $s'$  gives  $c, d$  with  $A \leq s' < s$  and  $B \leq H(x, y; s') = \min(x, T(s')) \leq T(s') < t$ .  $\square$

By the planar version of the strict widths lemma, Lemma 9.3 shows that  $\text{Fit}^\circ(a, b; u, v)$  holds if and only if the  $u \times v$  rectangle fits strictly into the  $a \times b$  rectangle.

**Proposition 9.4.** *If a check of  $\text{NC}^\circ(p, q)$  holds, then  $Q$  fits strictly into  $P$ .*

*Proof.* As in Proposition 4.1, let the check be  $\min(q_2, q_3) < p_1$  and  $\text{Fit}^\circ(p_2, p_3; q_1, h)$  with  $h = H^\circ(q_2, q_3; p_1) \geq 0$ . Lemma 9.3(a $^\circ$ ) for the rectangle  $q_1 \times h$  in the strip of width  $p_2$  gives  $c_2, d_2 \geq 0$ ,  $c_2^2 + d_2^2 = 1$ , with  $q_1 c_2 + h d_2 < p_2$  and  $q_1 d_2 + h c_2 < p_3$ . Since  $c_2, d_2 \leq 1$ , there is  $\eta > 0$  with

$$q_1 c_2 + (h + \eta) d_2 < p_2, \quad q_1 d_2 + (h + \eta) c_2 < p_3.$$

Lemma 9.3(a $^\circ$ ) for the rectangle  $q_2 \times q_3$  in the strip of width  $p_1$  with  $t = h + \eta$  gives  $c_1, d_1 \geq 0$ ,  $c_1^2 + d_1^2 = 1$ , with  $q_2 c_1 + q_3 d_1 < p_1$  and  $q_2 d_1 + q_3 c_1 < h + \eta$ . For the matrix  $R$  of Proposition 4.1 we obtain  $w_1 < p_1$ ,  $w_2 \leq q_1 c_2 + d_2(h + \eta) < p_2$  and  $w_3 \leq q_1 d_2 + c_2(h + \eta) < p_3$ .  $\square$

**Proposition 9.5.** *Let  $q \geq 0, p > 0$ , and let  $R \in \text{O}(3)$  satisfy  $w(R) < p$  and  $R_{ij} = 0$  for some  $i, j$ . Then  $\text{NC}^\circ(p, q)$  holds.*

*Proof.* Repeat the proof of Proposition 5.1 with  $c, d, c', d', B$  as there. From  $w_1 < p_1$  and Lemma 9.3(b $^\circ$ ),  $\min(q_2, q_3) < p_1$  and  $h := H^\circ(q_2, q_3; p_1) \leq B$ . Hence  $q_1 c' + d' h \leq w_2 < p_2$  and  $q_1 d' + c' h \leq w_3 < p_3$ , and Lemma 9.3(b $^\circ$ ) for the rectangle  $q_1 \times h$  in the strip of width  $p_2$  with direction  $(c', d')$  gives  $\min(q_1, h) < p_2$  and  $H^\circ(q_1, h; p_2) \leq q_1 d' + h c' < p_3$ , i.e.  $\text{Fit}^\circ(p_2, p_3; q_1, h)$ .  $\square$

*Proof of Theorem 9.2.* Sufficiency is Proposition 9.4. Conversely, let  $w(R) < p$ . With  $\varepsilon$  as in the proof of Theorem 9.1,  $Q$  fits into  $p - \varepsilon(1, 1, 1)$ , and Section 7 gives  $R' \in \text{O}(3)$  with a zero entry and  $w(R') \leq p - \varepsilon(1, 1, 1) < p$ . Proposition 9.5 concludes the proof.  $\square$

*Remark 9.6.* The change of the case split from  $x \leq s$  to  $x < s$  is essential:  $H(x, y; s)$  jumps at  $s = x$ . A face whose longer side equals the height may stand upright, touching the ceiling, with trace  $y$ ; if touching is forbidden, it cannot be tilted slightly (a small tilt increases its height), and its trace jumps to  $\min(x, T(x))$  with  $T(x) = y(3x^2 - y^2)/(x^2 + y^2)$ . For example, let  $p = (1, 1, 1)$  and  $q = (1, 0.5, 0.5)$ . The naive strict criterion (strict inequalities with  $H$ ) holds: with a horizontal edge 0.5, the face  $1 \times 0.5$  stands upright,  $h^* = H(1, 0.5; 1) = 0.5$ , and  $0.5 \times 0.5$  fits strictly into  $1 \times 1$ . But  $\text{NC}^\circ$  fails. Here  $T(1) = 0.5 \cdot 2.75/1.25 = 1.1$  for  $x = 1$ ,  $y = 0.5$ , so  $H^\circ(1, 0.5; 1) = \min(1, 1.1) = 1$ . If the horizontal edge is 1, then  $h = H^\circ(0.5, 0.5; 1) = 0.5$  and the floor requires  $H^\circ(1, 0.5; 1) < 1$ ; if it is 0.5, then  $h = H^\circ(1, 0.5; 1) = 1$  and the floor again requires  $H^\circ(0.5, 1; 1) < 1$ . Hence, by Theorem 9.2, this box does not fit strictly.

## 10. FORMAL VERIFICATION

The accompanying repository contains a Lean 4 formalization (Lean v4.34.0-rc2, Mathlib) of Theorems 1.2, 9.1 and 9.2, about 1800 lines in seven files:

- `Reduction.lean`, `Statement.lean`: definitions, Lemma 2.1 (via Mazur–Ulam), Lemma 3.1(a), Proposition 4.1;
- `ZeroEntry.lean`: Lemma 3.1(b) (by a chord argument instead of concavity), Proposition 5.1;
- `Motzkin.lean`: the theorem of the alternative (via Hahn–Banach separation);
- `Descent.lean`: Section 6;
- `Necessity.lean`: Section 7 and the main theorem `fits_iff_NC`;
- `Strict.lean`: Section 9, theorems `fitsStrict_iff_shrink` and `fitsStrict_iff_NC`s (with two further definitions, the open box and strict fitting).

The statement and the two definitions on which its meaning depends read (Unicode symbols of the source replaced by ASCII):

```
def box (q : Fin 3 -> Real) : Set E3 := {x | forall i, 0 <= x i /\ x i <= q i}
def Fits (p q : Fin 3 -> Real) : Prop :=
  exists f : IsometryEquiv E3 E3, f '' box q ⊆ box p
theorem fits_iff_NC (p q : Fin 3 -> Real) (hp : forall i, 0 < p i)
  (hq : forall i, 0 <= q i) : Fits p q <-> NC p q
```

Here  $E3$  is the Euclidean space  $\mathbb{R}^3$ , `box q` is the set of points  $x$  with  $0 \leq x_i \leq q_i$ , and `Fits p q` means that some isometry  $f$  of  $\mathbb{R}^3$  maps `box q` into `box p`. The criterion `NC` is Definition 1.1 verbatim (with indices 0, 1, 2). The command `#print axioms Boxes.fits_iff_NC` reports only the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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